

Lec 05: Introduction to Functional Programming

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Outline

- 1 Introduction Functional Programming
- 2 Monad
- 3 Expression system
- 4 Summary and Further Directions

Outline

1 Introduction Functional Programming

- Motivation
 - List: a good example
 - Nat and List
 - A little knowledge about termination
 - More Inductive Data Structure
 - Tree
 - Inductive Graph

2 Monad

- Monad Examples
- Monad Transformer

3 Expression system

Functional Programming: intro 1

- There are many programming languages having different advantages. An interesting kind of them is **functional programming** languages.
- Usually, in sequential programming language, we can define variables, data structures, functions and call them. However, this way is too messy. We can modify the value of variable at any location at any time.
- Is there some way to control the behavior of functions/methods in our programming language to make it more pure and has less side effect?

Functional Programming: Definition

Definition (Functional Programming)

Functional programming is a pattern for programming to make the function pure and has no side effect. Usually, the programming language implements algebraic data type (structure) and pattern match.

Here, the “definition” here is not strict mathematical definition, it's like a claim for new concept.

Functional Programming: intro 2

We can also consider the functional programming through the view of Curry-Howard (CH) corresponding:

- **Functional Programming.** We can consider the functional programming pattern is the corresponding of mathematical definition, which allow inductive structure.

Lean is a programming language, which means it has its programming pattern design and some type system; so the Functional Programming is very important.

Functional Programming: intro 3

Here is an example:

Example

There exist a function to calculate the factorial of input n , i.e.
 $\exists f, f : \mathbb{N} \rightarrow \mathbb{N}$ and $f(n) = n!$.

Proof.

It seems to be extreme easy, as the $n!$ could be defined through product, we just need to write down the function like:

$$f(n) := \begin{cases} 1 & \text{if } n = 0 \\ \prod_{i=1}^n i & \text{otherwise} \end{cases}$$

Thus, we can get the result directly. □

Functional Programming: intro 4

In this example, we may write the definition of this function in python like:

```
def f(n : int):  
    acc: int = 1  
    for i in range(n + 1):  
        acc = acc * i  
    return acc
```

- However, this function modify a variable, which is impure.
- **can you prove it in inductive way?**

Functional Programming: Inductive Proof

Proof.

Here is a proof written in inductive way: claim a function f to be:

$$f(n) := \begin{cases} 1 & \text{if } n = 0 \\ n \cdot f(n-1) & \text{otherwise} \end{cases}$$

To see the f has same result with $n!$.

- For $i = 0$, trivial;
- Assume f holds for $n = k$, for $k + 1$, we have:

$$f(k+1) = (k+1) \cdot f(k) = (k+1) \cdot k! = (k+1)!$$

Thus, f is the factorial function we need. □

Functional Programming: intro 5

After we have above inductive proof written in math, we can certainly write it in programming language allowing recursive definition like the following code in Lean:

```
def f : Nat -> Nat
:= fun n => match n with
  | Nat.zero => 1
  | Nat.succ k => (k + 1) * (f k)
```

Or in python:

```
def f (n : int):
    if (n == 0): return 1
    else: return n * f(n - 1)
```

You will see that it's really beautiful to write something in functional programming pattern.

Functional Programming: Takeaway

- It may be not a good example at beginning, the key here is the definition of $\mathbb{N}$ is based on inductive way. So we can handle it in inductive proof.
- Later, you will see that; for any data structure (type) written in inductive way, we can always do this magic things.
- Another good example I've learned is in one HoTT book (the proof of path induction).

Functional Programming: intro 6

Before we get start, to write a solid program, we need to discuss all the possible cases of input and make them used as we want. The core concepts are **constructor** and **eliminator**:

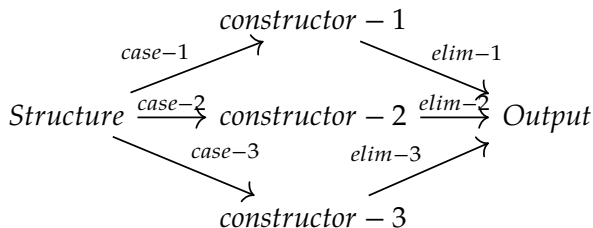
Definition

- **Constructor.** A type (data structure) is allowed to be defined through specified methods, who are the *constructors* of such a type;
- **Eliminator.** For a given type, if we want to define a function on it, we need to decomposite it by pattern matching, i.e. list all the possible cases and handle them one by one.

The two concepts is not so strict on distinguishing type theory and functional programming.

Functional Programming: intro final

Just like you discuss the possibility of a hypothesis to be case A or case B, we discuss the which the constructor is and handle them one by one:



Attention: the output need to be same type.

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4 Summary and Further Directions

List: intro

- In this section, we will learn the fundamental inductive structure, which is very similar with Peano natural number.
- I suggest you to do all the examples and exercises by your hand; all the high technicals are abstracted from such basic things.

List: definition

A list is defined as:

```
inductive List (a : Type u) where
| nil : List a
| cons (head : a) (tail : List a) : List a
```

recall that:

```
inductive Nat: Type where
| zero : Nat
| succ : Nat -> Nat
```

So, it's really similar to write some basic functions over list.

List: constructors

- Firstly, let try to define some new lists:

```
def lst1 : List Nat := []  
def lst2 : List Nat := [ 1 ]  
def lst3 : List Nat := List.cons 2 lst2
```

- I know you my be eager to write some real codes; but before we start, we need some necessary IO to show your result:

```
def main : IO ()  
:= do  
  ...  
  return ()
```

List Function 0: write some Nat function

Exercise

*Write the following functions on Nat using only pattern matching — no + or *:*

```
-- add 2 to any Nat (use .succ twice)
def add2 : Nat -> Nat := by admit
-- add two Nats (recursion on first argument)
def add : Nat -> Nat -> Nat := by admit
-- multiply two Nats (use add in the step)
def mul : Nat -> Nat -> Nat := by admit
```

Solution to the Exercise

Here is the solution:

```
def add2 : Nat -> Nat
:= fun n =>
  match n with
  | .zero => .succ .succ .zero
  | .succ k => .succ .succ .succ k
-- or def add2 := .succ .succ
def add : Nat -> Nat -> Nat
:= fun m n =>
  match m with
  | .zero => n
  | .succ k => match n with
    | .zero => m
    | .succ l => .succ (add k n)
```

Solution: Multiple Pattern Matching

You may use simultaneous pattern matching on both arguments:

```
def add : Nat -> Nat -> Nat
  := fun m n =>
    match m, n with
      | .zero, .zero    => .zero
      | .zero, .succ l => .succ l
      | .succ k, .zero => .succ k
      | .succ k, .succ l => .succ (.succ (add k l))
-- multiplication: m * (n+1) = m + m*n
def mul : Nat -> Nat -> Nat
  | _, .zero    => .zero
  | m, .succ k => add m (mul m k)
```

List Function 1: basic tools

Let's write some list function. Try to write the following functions on List a:

- head;
- length;
- take n (could return empty list if $n > \text{length}$);
- tail.

Solution to List Function 1

```

-- head returns Option: none when the list is empty
def head : List a -> Option a
| []      => none
| x :: _  => some x
-- length: count constructors
def length : List a -> Nat
| []      => 0
| _ :: xs => length xs + 1
-- take: stop at n = 0 or empty list
def take : Nat -> List a -> List a
| 0, _      => []
| _, []     => []
| n + 1, x :: xs => x :: take n xs
-- tail: drop the head, return the rest
def tail : List a -> List a
| []      => []
| _ :: xs => xs

```

List Function 2: apply some function

Assume in the following context, we only discuss the List over Nat.

Try to write:

- add one;
- multiple one;
- apply g.

```
def add1 : List Nat -> List Nat
| []      => []
| x :: xs => (x + 1) :: (add1 xs)
```

```
def mul2 : List Nat -> List Nat
| []      => []
| x :: xs => (x * 2) :: (mul2 xs)
```

```
def applyF (f : Nat -> Nat) : List Nat -> List Nat
| []      => []
| x :: xs => f x :: (applyF f xs)
```

All three follow the same recursive skeleton — `applyF` is the general pattern.

List Function 2: apply

This case can be abstracted as:

```
def map (f : a -> b) : List a -> List b
| []      => []
| x :: xs => f x :: map f xs
```

All the functions above are just use apply function over our list, we can define those function through the “map”.

List Function 3: acc

In this section, we try to handle two lists by induction.

- Reverse a list;
- Concat two lists;

This cases need to firstly define a new result container, like your first line in your sequential programming. This leads to the **acc**.

List Function 3: acc

We firstly check the solution:

```
def rev'
: List a -> List a -> List a
:= fun acc lst => match acc, lst with
  | acc, [] => acc
  | acc, x :: xs => rev' (x :: acc) xs

def rev : List a -> List a
:= rev' []
```

We can see that in the formal definition, we use the technical like sequential programming to firstly define some new variable as the container of result. If it's not so clear, let's firstly write them into one function:

List Function 3: acc

Here we use “let rec” to define a new recursive function.

```
def rev : List a -> List a
:= let rec loop : List a -> List a -> List a
    := fun acc lst => match acc, lst with
      | acc, [] => acc
      | acc, x :: xs => rev' (x :: acc) xs
  fun lst =>
    loop [] lst
```

The keyword is already recursive, but in the block of a function, we use pure functional pattern, we need to claim it's a self-calling function.

List Function 4: fold

Before we introduce the abstracted version of the reverse, we need to first see more easy case; let's write something like:

- Sum;
- Prod;

List Function 4: fold

We have the sum of list and product to be:

```
-- guards is a syntax sugar for pattern matching
def sumlst : List Nat -> Nat
:= let rec loop : List Nat -> Nat -> Nat
    | [], acc => acc
    | x :: xs, acc => loop xs (acc + x)
    fun lst => loop lst 0
#eval sumlst [ 1, 2, 3, 4 ] -- 10

def prodlst : List Nat -> Nat
:= let rec loop : List Nat -> Nat -> Nat
    | [], acc => acc
    | x :: xs, acc => loop xs (acc * x)
    fun lst => loop lst 1
#eval prodlst [ 1, 2, 3, 4 ] -- 24
```

List Function 4: foldr

We can abstract the pattern above:

```
def foldr (f : a -> b -> b) (init : b) : List a -> b
| []      => init
| x :: xs => f x (foldr f init xs)
```

- foldr processes from right to left, combining with f.
- foldl processes from left to right; this is the pattern in our rev function.

```
def sumlst' := List.foldr Nat.add 0
def prodlst' := List.foldr Nat.mul 1
#eval sumlst' [1, 2, 3, 4] -- 10
```

List Function 5: filter

Sometimes, we want to collect some elements on a list, it's the filter pattern:

- GE1, LE10;
- filter.

The pattern is the “filter”:

```
def filter : (a -> Bool) -> List a -> List a
:= fun p lst => match lst with
| [] => []
| x :: xs =>
  if p x
  then x :: (filter p xs)
  else filter p xs
```

Live Lean: `D5_ListFunctions.lean`

Essential List Functions in Lean 4

In Lean 4, all list utilities live in the `List` namespace. Functions that can fail return `Option` rather than panicking on empty input.

Function (Lean 4)	Description
<code>List.head? xs</code>	<code>Option α</code> — first element, none if empty
<code>List.getLast? xs</code>	<code>Option α</code> — last element, none if empty
<code>xs.length</code>	<code>Nat</code> — number of elements ($O(n)$)
<code>List.take n xs</code>	First n elements
<code>List.drop n xs</code>	All elements after the first n
<code>List.reverse xs</code>	Reversed list
<code>List.isEmpty xs</code>	<code>Bool</code> — true iff <code>[]</code>
<code>List.contains xs x</code>	Membership check: <code>Bool</code>

Essential List Functions in Lean 4 (Transform)

Function (Lean 4)	Description
<code>List.map f xs</code>	Apply <code>f</code> to every element
<code>List.filter p xs</code>	Keep elements where <code>p x = true</code>
<code>List.foldl f init xs</code>	Left fold: <code>f (f init x_1) x_2 ...</code>
<code>List.foldr f init xs</code>	Right fold: <code>f x_1 (f x_2 ...init)</code>
<code>List.flatMap xs f</code>	Map then flatten (<code>f : $\alpha \rightarrow \text{List } \beta$</code>)
<code>List.zip xs ys</code>	Pair up: <code>List ($\alpha \times \beta$)</code>
<code>List.zipWith f xs ys</code>	Combine element-wise with <code>f</code>
<code>List.join xss</code>	Flatten one level: <code>List (List α) $\rightarrow$ List α</code>

Essential List Functions in Lean 4 (Search & Sort)

Function (Lean 4)	Description
<code>List.find? p xs</code>	First element where <code>p x</code> : <code>Option α</code>
<code>List.findIdx? p xs</code>	Index of first match: <code>Option Nat</code>
<code>List.any p xs</code>	Any element satisfies <code>p</code> : <code>Bool</code>
<code>List.all p xs</code>	All elements satisfy <code>p</code> : <code>Bool</code>
<code>List.partition p xs</code>	Split into (passing, failing) pair
<code>List.sort xs</code>	Sort ascending (requires <code>Ord α</code>)
<code>List.sortBy cmp xs</code>	Sort with custom comparator
<code>List.dedup xs</code>	Remove consecutive duplicate elements

Note: Linked Lists and Performance

- The inductive `List` we have been using is a **singly-linked list**. Head access is $O(1)$, but random access and appending to the end are $O(n)$.
- In verified algorithm development, we first prove **correctness** using these simple inductive structures, then optimize performance separately.
- Lean 4's standard library also provides `Array` (backed by a contiguous heap array) for $O(1)$ random access when performance matters.

Exercise: Merge Sort in Lean 4

Implement merge sort. The key steps are:

```
-- Split a list into two halves
def splitHalf (xs : List Nat) : List Nat × List Nat :=
  (List.take (xs.length / 2) xs,
   List.drop (xs.length / 2) xs)

-- Merge two sorted lists into one sorted list
def merge : List Nat -> List Nat -> List Nat
| [], ys => ys
| xs, [] => xs
| x :: xs, y :: ys =>
  if x <= y then x :: merge xs (y :: ys)
  else y :: merge (x :: xs) ys

-- Merge sort (add termination_by xs.length for Lean)
def mergeSort (xs : List Nat) : List Nat :=
  if xs.length <= 1 then xs
  else let (l, r) := splitHalf xs
        merge (mergeSort l) (mergeSort r)
termination_by xs.length
```

1. Why Functions Must Terminate in Lean 4

Lean 4 is a **total functional language** rooted in dependent type theory. Every function must be proven to terminate in finite steps.

Logical Soundness (Curry-Howard Isomorphism)

In Lean, types are propositions and programs are proofs. If infinite recursion were allowed without verification, you could construct an infinite loop that produces a proof of False, rendering the mathematical system unsound.

```
-- If infinite loops were permitted:  
def proveFalse : False := proveFalse -- Logical collapse!
```

2. Structural vs. Well-Founded Recursion

Lean verifies termination using two distinct mechanisms:

Structural Recursion (Automatic)

- Peels off exactly one constructor per call ($x :: xs \rightarrow xs$).
- Lean automatically inspects the inductive type definition.
- No annotations needed.

Well-Founded Recursion

- Non-linear reductions (e.g., division, arithmetic, splitting lists).
- Automatic inspection cannot see that arguments shrink.
- **Requires annotation:** `termination_by`.

3. Syntax Pattern A: Direct Parameter Reference

When your function explicitly names its parameters in the header, write the decreasing expression directly after `termination_by`.

```
-- Euclidean algorithm for Greatest Common Divisor --  
def gcd (a b : Nat) : Nat :=  
  if b == 0 then a  
  else gcd b (a % b)  
termination_by b
```

Why this is legal: The expression `b` has type `Nat`. Because `a % b < b` for any `b > 0`, Lean verifies that the natural number strictly decreases toward 0.

4. Syntax Pattern B: Explicit Variable Binding

When pattern matching anonymously across multiple lines, you must bind the parameter names before the arrow ($\Rightarrow$) so the measure expression can reference them.

```
def merge : List Int -> List Int -> List Int
| [], ys => ys
| xs, [] => xs
| x::xs, y::ys =>
  if x <= y then x :: merge xs (y::ys)
  else y :: merge (x::xs) ys
termination_by xs ys => xs.length + ys.length
```

Measure: The sum `xs.length + ys.length` strictly drops by exactly 1 on every recursive branch.

5. Advanced: Lexicographical Tuples

For complex multi-variable recursion where one variable might increase while another decreases, use a **tuple expression** (x, y) .

```
-- Ackermann function requires lexicographical order --  
def ack : Nat -> Nat -> Nat  
  | 0,   y   => y + 1  
  | x+1, 0   => ack x 1  
  | x+1, y+1 => ack x (ack (x + 1) y)  
termination_by x y => (x, y)
```

Well-Founded Relation: Lean compares tuples from left to right: either x strictly shrinks, or x stays identical and y strictly shrinks.

6. Manual Proofs with decreasing_by

If Lean's automatic tactics cannot prove your measure decreases, attach a `decreasing_by` block immediately following `termination_by`.

```
def div2 (n : Nat) : Nat :=  
  if h : n < 2 then 0  
  else 1 + div2 (n - 2)  
termination_by n  
decreasing_by  
  -- Provide explicit proof that  $n - 2 < n$   
  omega
```

7. Summary of Legal Measure Types

Measure Type	Example Syntax	Typical Use Case
Nat	<code>termination_by n</code>	Countdown loops, arithmetic steps
List Length	<code>termination_by xs.length</code>	Custom list splitting / slicing
Combined Sum	<code>... => xs.length + ys.length</code>	Interleaved multi-list processing
Tuple (Lex)	<code>... x y => (x, y)</code>	Nested recursive calls (Ackermann)
Custom Proof	<code>decreasing_by exact ...</code>	Non-standard arithmetic bounds

Interesting Project: Sudoku in Lean 4

Sudoku is a classic exercise for functional programmers:

- **Grid as data:** represent a 9×9 grid as `Array (Array (Option Nat))`
- **Constraint checking:** write `validRow`, `validCol`, `validBox` using `List.nodup`
- **Backtracking solver:** recursive search with `Option` for failure — a natural fit for monadic style
- **Proof challenge:** prove that your checker is complete (never rejects a valid board)

Not required, but highly recommended if you want hands-on FP practice beyond the exercises.

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Tree definition

We can define tree as:

```
inductive Tree := Nil | Node Tree Tree
```

But it carry no information

More Trees

We modify our tree to:

```
inductive Tree a :=  
  Leaf a  
| Node (Tree a) a (Tree a)
```

Exercises

Write the following tree functions:

- size; depth; reflect: swap left and right.
- isBST, balanced;
- preorder, inorder, postorder.

Implement:

- insert;
- search;

Problem

Write a binary balanced tree.

- A cheating implementation: from sortedlist
- Regular implementation.

Inductive Graph: Three Representations

Graphs can also be defined inductively! Three common approaches:

- **Algebraic Graph** (Mokhov): build graphs via constructors

```
inductive Graph (a : Type) where
  | empty    : Graph a
  | vertex   : a -> Graph a
  | overlay  : Graph a -> Graph a -> Graph a -- union
  | connect  : Graph a -> Graph a -> Graph a -- add all edges
```

- **Edge list**: $\text{List } (a \times a)$ — pairs of connected vertices
- **Adjacency list**: $\text{List } (a \times \text{List } a)$ — vertex $\rightarrow$ neighbour list

Exercises: Graph Functions

Using the algebraic graph representation, write:

- `vertexCount : Graph a -> Nat` — number of distinct vertices
- `hasEdge : Graph Nat -> Nat -> Nat -> Bool` — edge membership
- `toEdgeList : Graph a -> List (a × a)` — extract all edges from `connect`
- **Bonus:** prove `overlay g empty = g` for any graph `g`

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Monad intro 1: why we need Monad?

- Last section, we share the core of functional programming. This section, we introduce the most difficult and important concept: the “Monad”.
- For simplification, one can think of a monad as a pattern used in functional programming to sequence computations that share a common context or effect.

Let's firstly see some examples.

Monad intro 1: motivation from side effect handle

- In previous context, we have the function “take n” on list. But it could failed, so we can try to make a new algebraic data structure to avoid this:

```
inductive Optional (a : Type) : Type where  
| none : Optional a  
| some : a -> Optional a
```

- Use this, we can define the “optional take n”. But it’s nearly same with the classical one because we already use nil as empty return.
- A good idea is to return some string or error type to represent the information.

Monad intro 2: side effect 2

We define the “Either” type:

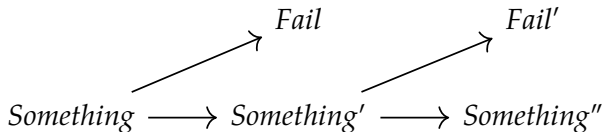
```
inductive Either (a b : Type) : Type where
| left  : a -> Either a b
| right : b -> Either a b
```

We can define take n to be:

```
def Etaken
: Nat -> List Nat -> Either Nat String
:= fun n lst => match lst with
| [] => Either.right "Empty List"
| x :: xs =>
    if n == 0
    then Either.left x
    else Etaken (n - 1) xs
```


Monad intro 3: abstract the framework of Monad

The examples show a great idea:



If we abstract a structure to include the “Something” and “Fail”, we actually do the following things.

Monad intro 3: Monad bind

The procedure we need to define is:

$$Container(x) \multimap \longrightarrow Container(x)$$

We extract the inner and create a new container, it's formally defined as the bind in Monad.

The Inventors of Monads



Eugenio Moggi (1960–)

- Italian computer scientist
- *“Notions of Computation and Monads”* (1991) — introduced monads as a semantic framework for computational effects



Philip Wadler (1956–)

- *“Comprehending Monads”* (1992) — brought monads into practical FP (Haskell)
- Co-designer of Haskell; also proved Curry-Howard holds for System F

Monad intro 4: formal definition

Formally, a **Monad** on a type universe is a type constructor $m : \text{Type } u \rightarrow \text{Type } v$ equipped with two fundamental operations:

```
class Monad (m : Type u -> Type v) where
  pure  : a -> m a
  bind  : m a -> (a -> m b) -> m b
```

- **pure** (or *return*): Embeds a clean value into the monadic effect container.
- **bind** ($\gg$): Chains a computation that produces an effect onto an existing effectful value.

Monad intro 5: syntactic sugar (do-notation)

Writing nested bind functions manually becomes unreadable. Lean provides **do-notation** as syntactic sugar for sequential pipelines:

```
-- Raw Bind:
def calc : Option Nat :=
  findX >>= fun x =>
    findY >>= fun y =>
      pure (x + y)
-- do notation:
def calc : Option Nat := do
  let x <- findX
  let y <- findY
  return (x + y)
```

Under the hood, Lean desugars the `<-` arrow directly into bind.

Live Lean: `D5_Monad.lean`

IO Monad: interacting with the real world

Pure functions cannot interact with the outside world (printing to console, reading files) because these operations have side effects.

- The **IO** monad wraps computation steps that interact with the external environment.
- A value of type `IO a` represents a description of a computation that, when executed by the runtime, produces a value of type `a`.

```
def greet : IO Unit := do
  IO.println "What is your name?"
  let name <- IO.getLine
  IO.println s!"Hello, {name}!"
```

IO Monad: controlled mutability with IORef

In functional programming, variables are immutable by default. But real systems (caches, counters, UI state) require heap mutability.

Enter `IORef`:

- An `IORef a` is a reference to a heap-allocated mutable cell holding a value of type `a`.
- Reading and mutating it are explicitly tracked as `IO` effects, keeping the rest of your math logic referentially transparent!

IO Monad: Controlled Mutability (IO.Ref)

While functional data is immutable, real systems require heap mutability. Lean provides `IO.Ref` for safe, reference-counted cell mutation.

Here are the api in lean:

```
IO.mkRef {a : Type} (a : a) : IO (IO.Ref a)
IO.Ref.get (self : IO.Ref a) : IO a
IO.Ref.modify (self : IO.Ref a) (f : a -> a) : IO Unit

def counterDemo : IO Unit := do
  let r <- IO.mkRef 0
  r.modify (fun n => n + 1) -- In-place C update if RC=1
  let val <- r.get
  IO.println s!"Counter value: {val}"
```


IO Monad: native exception handling

Real-world operations fail (e.g., missing files, network timeouts). In Lean 4, `IO a` is secretly an alias for `EIO IO.Error a`—meaning exception handling is baked directly into the monad!

- You can use structured `try ... catch ... finally` blocks inside `do`-notation just like imperative languages.

```
def readConfigSafely (path : String) : IO String := do
  try
    -- IO.FS handles raw filesystem streams
    let content <- IO.FS.readFile path
    return content
  catch e =>
    -- e has type IO.Error
    IO.eprintln s!"[Warning] Could not load {path}: {e}"
    return "DEFAULT_CONFIG_DATA"
```

What makes Lean 4's IO special?

Because Lean 4 compiles directly to optimized C code, its `IO` monad possesses two systems-level superpowers:

- **Lightweight Parallelism** (`Task`): You can spawn background OS threads effortlessly using `IO.asTask`. Lean's runtime handles work-stealing and memory synchronization automatically.
- **Zero-Overhead C FFI**: You can bind raw C/Rust functions directly into Lean's `IO` monad using the `@[extern]` attribute.

```
def parallelCrunch : IO Unit := do
  -- Spawn two heavy calculations on background threads
  let task1 <-
    IO.asTask (pure (heavyPrimeCalc 100000))
  let task2 <-
    IO.asTask (pure (heavyPrimeCalc 200000))

  -- Block until both tasks finish execution
  let res1 <- IO.ofExcept task1.get
  let res2 <- IO.ofExcept task2.get
  IO.println s!"Combined: {res1 + res2}"
```

The Abstraction Hierarchy: Functor & Applicative

In Lean 4, a **Monad** is not an isolated concept—it is the top of a strict typeclass hierarchy: $\text{Functor} \rightarrow \text{Applicative} \rightarrow \text{Monad}$.

```
class Functor (f : Type u -> Type v) where
  -- <$>
  map : {a b : Type u} -> (a -> b) -> f a -> f b
class Applicative (f : Type u -> Type v)
  extends Functor f where
  pure : {a : Type u} -> a -> f a
  -- <*>
  seq  : {a b : Type u} -> f (a -> b) -> (Unit -> f a) -> f b
```

- **Functor** (`<$>`): Applies a pure function inside an effect container.
- **Applicative** (`<*>`): Applies an *effectful function* to an effectful argument (allowing independent, static computation trees).

Why Applicative before Monad?

Why not jump straight to `Monad`? Because `Applicative` handles **independent effects** cleanly without requiring full sequential evaluation chains.

```
-- Evaluates arguments independently
def addOpt (x y : Option Nat)
  : Option Nat :=
  Pure.pure Nat.add <*> x <*> y
-- Sequential / dynamic branching
def addOpt (x y : Option Nat)
  : Option Nat := do
  let a <- x
  let b <- y
  return (a + b)
```

Key Difference: With `Monad`, the next computation can depend on the *value* of the previous one. With `Applicative`, the structure of computation is static.

Outline

- 1 Introduction Functional Programming
 - Nat and List
 - A little knowledge about termination
 - Tree
 - Inductive Graph
- 2 **Monad**
 - Monad: introduction
 - **Monad examples and Monad Transformer**
 - Monad Examples
 - Monad Transformer
- 3 Expression system
- 4 Summary and Further Directions

Important Monad Examples

Beyond `Option` and `IO`, real programs use several standard monads:

- **Reader** — read-only shared environment (e.g. configuration)
- **Writer** — accumulate output (e.g. logging, diagnostics)
- **State** — thread mutable state through pure code (e.g. counters, accumulators)
- **Monad Transformer** (`ReaderT`, `StateT`, ...) — stack two monads into one

Each is just a type with `pure` and `bind` — the structure you already know.

Reader, Writer, and State Monads

Beyond error handling and IO, monads simulate stateful programs purely:

- **Reader Monad** (`Reader r a`): Passes a read-only environment `r` down a computation tree implicitly (like global config).
- **Writer Monad** (`Writer w a`): Accumulates secondary outputs (like execution logs) alongside computation.
- **State Monad** (`State s a`): Threads a mutable state `s` through pure functions.

```
-- State s a is fundamentally a wrapper around:  
-- s -> (a, s)
```


The Reader Monad (ReaderM)

The **Reader Monad** passes a read-only environment (like global configurations or database connections) implicitly down a computation tree.

```
abbrev ReaderM (rho : Type u) (a : Type v) := rho -> a
read : ReaderM rho rho -- Extracts the shared environment

structure AppConfig where
  dbHost : String
  port   : Nat

def buildConnectionUrl : ReaderM AppConfig String := do
  let cfg <- read -- Implicitly fetches AppConfig
  return s!"http://{cfg.dbHost}:{cfg.port}/api"
```

The State Monad (StateM)

The **State Monad** threads mutable state purely through functional code by secretly transforming functions into state-transition pairs:

$\sigma \rightarrow \alpha \times \sigma$.

```
abbrev StateM (s : Type u) (a : Type v)
:= s -> a × s
get : StateM s s
set (s : s) : StateM s PUnit
modify (f : s -> s) : StateM s PUnit
```

The State Monad (StateM)

```
def incrementAndGet : StateM Nat Nat := do  
  modify (fun s => s + 1)  
  get
```

The Writer Monad Pattern (Logging)

The **Writer Monad** accumulates secondary outputs (like compiler diagnostics or execution logs) alongside primary computations without polluting return types.

Idiomatic Lean 4 Abstraction

```
-- In Lean 4, Writer is idiomatically
-- modeled via append-only State
abbrev WriterM (w : Type) := StateM (Array w)
def tell (output : w) : WriterM w Unit
:= modify (·.push output)
```

The Writer Monad Pattern (Logging)

```
def divideWithLog (a b : Nat)
: WriterM String Nat := do
  if b == 0 then
    tell "Error: Division by zero attempted!"
    return 0
  else
    tell s!"Dividing {a} by {b}"
    return a / b
```

Monad Transformers: stacking effects

What if we want both error handling (`Except`) *and* state (`State`)? We use **Monad Transformers** (ending in `T`).

Base Monad (`IO`) $\xrightarrow{\text{StateT}}$ `StateT Env IO a`

```
-- Wraps IO to also carry a symbol table state:
abbrev CompilerM := StateT SymbolTable IO
```

Transformers allow us to build custom domain-specific languages (DSLs) by layering capabilities cleanly.

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Expression Language: A Capstone Example

Implementing a mini programming language ties together everything from this course:

- **Inductive types** — define syntax (AST) as a recursive data structure
- **Pattern matching** — evaluate expressions by case analysis on constructors
- **Option monad** — handle variable lookup failures cleanly with do-notation
- **Proof by induction** — verify that the evaluator satisfies algebraic laws

The same pipeline (AST $\rightarrow$ evaluator $\rightarrow$ prover) scales to real language implementations.

Abstract Syntax Trees (AST)

To implement a small language, we firstly model its syntax using an inductive type. Let's build a simple arithmetic expression language:

```
inductive Expr where
  | val   : Nat -> Expr
  | var   : String -> Expr
  | add   : Expr -> Expr -> Expr
  | mul   : Expr -> Expr -> Expr
  deriving Repr
```

An expression like $(x + 2) \times 3$ is represented as:

```
def myExpr : Expr :=
  Expr.mul (Expr.add (Expr.var "x") (Expr.val 2))
           (Expr.val 3)
```

Evaluating Expressions: An Environment

To evaluate expressions containing variables (`Expr.var`), we need an **Environment** (a symbol table) mapping variable names to numerical values.

```
abbrev Env := String -> Option Nat
def emptyEnv : Env := fun _ => none
def extendEnv (k : String) (v : Nat) (env : Env)
: Env :=
  fun name =>
    if name == k
    then some v
    else env name
```

Because lookup might fail for undefined variables, our evaluator must return an `Option Nat`.

The Evaluator (Pattern Matching & Monads)

We write the evaluator by induction on the Expr structure. Notice how cleanly do-notation handles potential lookup failures:

```
def eval (e : Expr) (env : Env)
: Option Nat := match e with
| Expr.val n => pure n
| Expr.var s => env s
| Expr.add l r => do
  let vl <- eval l env
  let vr <- eval r env
  return (vl + vr)
| Expr.mul l r => do
  let vl <- eval l env
  let vr <- eval r env
  return (vl * vr)
```

Live Lean: D5_Expr.lean

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- Meta-Programming

Lean's Expr

Lean is selfhosted, you can write functional programs in Lean that analyze, generate, and prove properties about Lean expressions directly!

```
-- Expr:
inductive Lean.Expr where
| bvar (deBruijnIndex : Nat)
| fvar (fvarId : FVarId)
| const (declName : Name)
    (us : List Level)
| app (fn : Expr) (arg : Expr)
| lam (binderName : Name)
    (ty : Expr) (body : Expr)
```

Syntax and Evaluation: The Pipeline

- **Syntax** (BNF $\leftrightarrow$ inductive type): each production rule maps directly to a constructor.

$$e ::= n \mid x \mid e_1 + e_2 \mid e_1 \times e_2$$

corresponds exactly to `val`, `var`, `add`, `mul`.

- **Evaluation** (pattern match on constructors): each constructor has one case in `eval`, making the correspondence between syntax and semantics structurally explicit.
- **Failure propagation**: variable lookup returns `Option Nat`; the monad threads failure automatically through the recursive cases.

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Project: Build a Mini Language in Lean 4

A classic exercise: extend the Expr language into a real mini-interpreter.

- 1 **Extend the AST:** add `ite` (if-then-else), `letIn` (local binding), `lam/app` (lambda & apply)
- 2 **Extend the evaluator:** handle closures using a value type `Val` (`VNum`, `VBool`, `VClos`)
- 3 **Add a parser:** write a recursive-descent parser from strings to `Expr`
- 4 **Prove correctness:** show your evaluator respects substitution laws

Inspiration: “Write Yourself a Scheme in 48 Hours” (classic Haskell tutorial at wikibooks.org) — port it to Lean 4!

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Meta-Programming in Lean 4

Lean 4 is also its own meta-language: you can extend the syntax from within.

- **Macro system** (`macro`, `syntax`): define new surface syntax that expands into existing Lean terms at parse time — no runtime cost.
- **Elaboration** (`elab`): run arbitrary Lean code during type-checking to construct terms, perform tactics, or generate definitions.
- **Lean.Expr**: the internal AST that represents all Lean terms — the same inductive tree structure you built above, but for Lean itself.

Bottom line: the Expr language you built is a microcosm of how Lean itself works internally.

Architectural Trade-offs: Extrinsic vs. Intrinsic

Dimension	Extrinsic Proofs	Intrinsic Proofs

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Industry Best Practice: *"Extrinsic internally, Intrinsic at the boundary."* Use Extrinsic for private helper functions, but wrap public, security-critical APIs in Subtypes to provide an absolute safety contract.

Metaprogramming in Lean 4

Metaprogramming in Lean 4

Syntax, Macros, and the Elaboration System

The Lean 4 Paradigm: Complete Reflection

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- This means the "meta-level" is reflected down to the "object-level". You can define a new piece of syntax, write a tactic to handle it, and use it in the very next line of the same file.
- For large verification projects, this allows us to build **Domain Specific Languages (DSLs)** that hide boilerplate and automate proofs.

The Compilation Pipeline

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To write a custom automated prover, you must understand how Lean translates a string of characters into a mathematically verified theorem.

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- ➍ **Kernel Checking ($\text{Expr} \rightarrow \text{Verified}$):** The highly trusted, tiny core of Lean checks the Expr against the foundational axioms.

Step 1: Extending the Parser (syntax)

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Before we can evaluate custom code, we must teach the Lean parser to recognize it. We do this using the `syntax` command.

```
-- Define a new syntax category for our project if needed
declare_syntax_cat my_cmds

-- Inject new syntax into the global `term` category
-- `syntax` [precedence] [elements] `:` [category]
syntax:10 term:10 " XOR " term:11 : term

-- We can also define complex project-specific blocks
syntax "BEGIN_VERIFY " ident " EXPECT " num " END" : command
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- Lean's parser is an extensible Pratt parser.
- If you stop here and type `true XOR false`, Lean will throw an error: *"elaboration function not found for..."* because the parser recognizes it, but the compiler doesn't know what it means yet.

Step 2: Macros (Syntax-to-Syntax)

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- **Quotations (```):** We use the backtick to construct syntax trees.
- **Antiquotations (`$`):** We use the dollar sign to inject captured variables into those syntax trees.

Writing a Macro

Writing a Macro

Let's write the macro rule for our XOR operator.

```
-- Tell Lean how to rewrite the syntax tree
macro_rules
| `($l XOR $r) => `((Not $l && $r) || ($l && Not $r))

-- Usage
#eval true XOR false -- Output: true
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-- Usage
#eval true XOR false -- Output: true
```

Lean provides a shortcut macro command that combines syntax and macro_rules:

```
macro "ASSERT_EQUAL " a:term " AND " b:term : tactic =>
  `(tactic| exact rfl)
```

Step 3: The Elaborator (Elab)

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When macros bottom out or you need to inspect types, the **Elaborator** takes over. Its job is to bridge the gap between human-readable text and kernel-readable math.

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- **The Input:** Untyped Syntax.
- **The Output:** A fully typed Expr (Expression) or a modified State.
- **The Work:** To build an Expr, the Elaborator must resolve implicit arguments, synthesize typeclasses, unfold definitions, and solve unification problems using Metavariables (holes).

The Three Flavors of Elaboration Monads

Depending on what your project is building, Lean places you in a specific Monadic context with different capabilities:

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- 1 **CommandElabM**: Operates at the top level of a file. Used to build commands like `#check`, `def`, or custom environment modifiers. Returns `Unit`.

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Depending on what your project is building, Lean places you in a specific Monadic context with different capabilities:

- 1 **CommandElabM**: Operates at the top level of a file. Used to build commands like `#check`, `def`, or custom environment modifiers. Returns `Unit`.
- 2 **TermElabM**: Operates inside definitions. Takes a Syntax tree and an expected type, and attempts to return an `Expr`.

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Depending on what your project is building, Lean places you in a specific Monadic context with different capabilities:

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- 2 **TermElabM**: Operates inside definitions. Takes a Syntax tree and an expected type, and attempts to return an `Expr`.
- 3 **TacticM**: Operates inside a `by` block. It manages a list of open Metavariables (the "Goals") and applies transformations to close them.

Writing a Command Elaborator

Writing a Command Elaborator

Let's write a custom top-level command that just prints "Hello" to the compiler trace. We use the `elab` shortcut.

```
open Lean Elab Command
```

```
-- Define the syntax and bind it directly to CommandElabM
```

```
elab "#hello" : command => do  
  logInfo "Hello from the Elaborator!"
```

```
-- Invoke it
```

```
#hello
```

```
-- Lean compiler outputs: Hello from the Elaborator!
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Note: The `elab` keyword automatically handles registering the syntax and wrapping your code in the correct macro-rule handlers.

Writing a Tactic Elaborator: Interacting with State

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Tactics are just elaborators that run in `TacticM`. Let's write a tactic that inspects the current goal and prints its type.

```
open Lean Elab Tactic

elab "print_goal" : tactic => do
  -- 1. Get the primary metavariable we are trying to solve
  let mvarId <- getMainGoal

  -- 2. Get the declaration (which contains the type)
  let decl <- mvarId.getDecl

  -- 3. Log the type as formatted MessageData (m!"...")
  logInfo m!"The current goal type is: {decl.type}"
```

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  -- 3. Log the type as formatted MessageData (m!"...")
  logInfo m!"The current goal type is: {decl.type}"
```

In a large project, you will use these primitives to inspect the `Expr` tree of the goal, decide which theorems to apply, and build a custom automated solver.

The Ultimate Cheat Code: evalTactic

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You do not always have to manually manipulate Expr trees (which is extremely difficult). You can pass variables from the Syntax tree directly back into the tactic engine.

```
elab "custom_intro " name:ident : tactic => do  
  
  evalTactic <- `(tactic| intro $name)  
  
  evalTactic <- `(tactic| simp)
```

- This demonstrates the power of Lean 4: you can write tactics using raw AST manipulation, OR you can recursively call the syntax engine inside the tactic using evalTactic.
- This is how you will build the "Automated Complexity Prover" for your final project.

Summary: Macro vs. Elab

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When designing the architecture of a major Lean project, how do you choose?

Feature	Macros	Elaborators (elab)
I/O	Syntax -> Syntax	Syntax -> Expr / State
Type Aware	Blind	Fully Type-Aware
Complexity	Very Low (Pattern Match)	High (Monadic API)
Use Case	Syntactic Sugar, Notation	Custom Tactics, Inference

Golden Rule for Big Projects: Always try to implement a feature as a Macro first. Only drop down into Elab if you absolutely must inspect the types of variables to decide what to do next.

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What We Covered Today

Part 1 — Functional Programming

- Pure functions, immutability, pattern matching
- Recursive types: Nat, List, Tree
- Higher-order functions: map, filter, fold
- Termination: structural vs. well-founded recursion

Part 2 — Monads

- Functor / Applicative / Monad

Part 3 — Expression Languages

- Syntax as an inductive type
- Evaluation pipeline: parse $\rightarrow$ eval
- Typed vs. untyped interpreters
- Meta-programming with Syntax and Macro

Key Insight

- A Lean program *is* a proof term
- Monads generalize

The Hierarchy We Have Built

Layer	Abstraction	Lean typeclass
Data	Inductive types	(built-in)
Transformation	Pure functions	Function
Mapping	Structure-preserving map	Functor
Sequencing	Context-aware chaining	Monad
Effects	I/O, state, error	IO, StateT, ...
Language	Syntax + semantics	Macro, Elab

Each layer builds on the one below. Lean's type system ensures the connections are consistent.

Connections to the Rest of the Course

- **D3 (Typeclasses)** — Functor, Monad are typeclass instances; today's code *is* the typeclass system in action.

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- **D3 (Typeclasses)** — `Functor`, `Monad` are typeclass instances; today's code *is* the typeclass system in action.
- **D4 (Analysis / Topology)** — `Filter` in `Mathlib` is a monad-like structure; limits are encoded with `Tendsto` using filter functors.

Connections to the Rest of the Course

- **D3 (Typeclasses)** — `Functor`, `Monad` are typeclass instances; today's code *is* the typeclass system in action.
- **D4 (Analysis / Topology)** — `Filter` in `Mathlib` is a monad-like structure; limits are encoded with `Tendsto` using filter functors.
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- **Meta-programming** — `Macro` and `Elab` (Part 3) are the foundation for writing custom tactics and notation in Lean 4.

Further Reading and Projects

Recommended Reading

- *Functional Programming in Lean*
(David Thrane Christiansen, free online)
- *Lean 4 Metaprogramming*
(Arthur Paulino et al., free online)
- *Haskell Programming from First Principles*
— Monad chapters transfer directly to Lean

Project Ideas

- Implement a **JSON parser** using Parsec-style monads
- Write a **Sudoku solver** with backtracking in StateT
- Build a **tiny calculator** with a typed expression language and a verified evaluator
- Prove **monad laws** for your custom monad
- Implement **Stream** (lazy lists) using coinductive

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- Lean is the rare language that lets you *prove* this equivalence, not just believe it.